

2 Cauchy's Theorem and Its Applications

The solution of a large number of problems can be reduced, in the last analysis, to the evaluation of definite integrals; thus mathematicians have been much occupied with this task... However, among many results obtained, a number were initially discovered by the aid of a type of induction based on the passage from real to imaginary. Often passage of this kind led directly to remarkable results. Nevertheless this part of the theory, as has been observed by Laplace, is subject to various difficulties...

After having reflected on this subject and brought together various results mentioned above, I hope to establish the passage from the real to the imaginary based on a direct and rigorous analysis; my researches have thus led me to the method which is the object of this memoir...

A. L. Cauchy, 1827

In the previous chapter, we discussed several preliminary ideas in complex analysis: open sets in $\mathbb{C}$, holomorphic functions, and integration along curves. The first remarkable result of the theory exhibits a deep connection between these notions. Loosely stated, Cauchy's theorem says that if f is holomorphic in an open set Ω and $\gamma \subset \Omega$ is a closed curve whose interior is also contained in Ω then

$$(1) \quad \int_{\gamma} f(z) dz = 0.$$

Many results that follow, and in particular the calculus of residues, are related in one way or another to this fact.

A precise and general formulation of Cauchy's theorem requires defining unambiguously the “interior” of a curve, and this is not always an easy task. At this early stage of our study, we shall make use of the device of limiting ourselves to regions whose boundaries are curves that are “toy contours.” As the name suggests, these are closed curves whose visualization is so simple that the notion of their interior will be unam-

biguous, and the proof of Cauchy's theorem in this setting will be quite direct. For many applications, it will suffice to restrict ourselves to these types of curves. At a later stage, we take up the questions related to more general curves, their interiors, and the extended form of Cauchy's theorem.

Our initial version of Cauchy's theorem begins with the observation that it suffices that f have a primitive in Ω , by Corollary 3.3 in Chapter 1. The existence of such a primitive for toy contours will follow from a theorem of Goursat (which is itself a simple special case)¹ that asserts that if f is holomorphic in an open set that contains a triangle T and its interior, then

$$\int_T f(z) dz = 0.$$

It is noteworthy that this simple case of Cauchy's theorem suffices to prove some of its more complicated versions. From there, we can prove the existence of primitives in the interior of some simple regions, and therefore prove Cauchy's theorem in that setting. As a first application of this viewpoint, we evaluate several real integrals by using appropriate toy contours.

The above ideas also lead us to a central result of this chapter, the Cauchy integral formula; this states that if f is holomorphic in an open set containing a circle C and its interior, then for all z inside C ,

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Differentiation of this identity yields other integral formulas, and in particular we obtain the regularity of holomorphic functions. This is remarkable, since holomorphicity assumed only the existence of the first derivative, and yet we obtain as a consequence the existence of derivatives of all orders. (An analogous statement is decisively false in the case of real variables!)

The theory developed up to that point already has a number of noteworthy consequences:

- The property at the base of “analytic continuation,” namely that a holomorphic function is determined by its restriction to any open subset of its domain of definition. This is a consequence of the fact that holomorphic functions have power series expansions.

¹Goursat's result came after Cauchy's theorem, and its interest is the technical fact that its proof requires only the existence of the complex derivative at each point, and not its continuity. For the earlier proof, see Exercise 5.

- Liouville's theorem, which yields a quick proof of the fundamental theorem of algebra.
- Morera's theorem, which gives a simple integral characterization of holomorphic functions, and shows that these functions are preserved under uniform limits.

1 Goursat's theorem

Corollary 3.3 in the previous chapter says that if f has a primitive in an open set Ω , then

$$\int_{\gamma} f(z) dz = 0$$

for any closed curve γ in Ω . Conversely, if we can show that the above relation holds for some types of curves γ , then a primitive will exist. Our starting point is Goursat's theorem, from which in effect we shall deduce most of the other results in this chapter.

Theorem 1.1 *If Ω is an open set in $\mathbb{C}$, and $T \subset \Omega$ a triangle whose interior is also contained in Ω , then*

$$\int_T f(z) dz = 0$$

whenever f is holomorphic in Ω .

Proof. We call $T^{(0)}$ our original triangle (with a fixed orientation which we choose to be positive), and let $d^{(0)}$ and $p^{(0)}$ denote the diameter and perimeter of $T^{(0)}$, respectively. The first step in our construction consists of bisecting each side of the triangle and connecting the mid-points. This creates four new smaller triangles, denoted $T_1^{(1)}, T_2^{(1)}, T_3^{(1)}$, and $T_4^{(1)}$ that are similar to the original triangle. The construction and orientation of each triangle are illustrated in Figure 1. The orientation is chosen to be consistent with that of the original triangle, and so after cancellations arising from integrating over the same side in two opposite directions, we have

$$(2) \quad \int_{T^{(0)}} f(z) dz = \int_{T_1^{(1)}} f(z) dz + \int_{T_2^{(1)}} f(z) dz + \int_{T_3^{(1)}} f(z) dz + \int_{T_4^{(1)}} f(z) dz.$$

For some j we must have

$$\left| \int_{T^{(0)}} f(z) dz \right| \leq 4 \left| \int_{T_j^{(1)}} f(z) dz \right|,$$

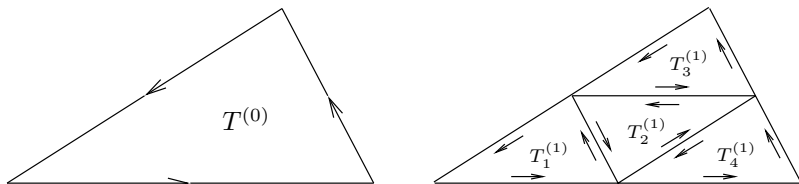


Figure 1. Bisection of $T^{(0)}$

for otherwise (2) would be contradicted. We choose a triangle that satisfies this inequality, and rename it $T^{(1)}$. Observe that if $d^{(1)}$ and $p^{(1)}$ denote the diameter and perimeter of $T^{(1)}$, respectively, then $d^{(1)} = (1/2)d^{(0)}$ and $p^{(1)} = (1/2)p^{(0)}$. We now repeat this process for the triangle $T^{(1)}$, bisecting it into four smaller triangles. Continuing this process, we obtain a sequence of triangles

$$T^{(0)}, T^{(1)}, \dots, T^{(n)}, \dots$$

with the properties that

$$\left| \int_{T^{(0)}} f(z) dz \right| \leq 4^n \left| \int_{T^{(n)}} f(z) dz \right|$$

and

$$d^{(n)} = 2^{-n}d^{(0)}, \quad p^{(n)} = 2^{-n}p^{(0)}$$

where $d^{(n)}$ and $p^{(n)}$ denote the diameter and perimeter of $T^{(n)}$, respectively. We also denote by $\mathcal{T}^{(n)}$ the *solid* closed triangle with boundary $T^{(n)}$, and observe that our construction yields a sequence of nested compact sets

$$\mathcal{T}^{(0)} \supset \mathcal{T}^{(1)} \supset \dots \supset \mathcal{T}^{(n)} \supset \dots$$

whose diameter goes to 0. By Proposition 1.4 in Chapter 1, there exists a unique point z_0 that belongs to all the solid triangles $\mathcal{T}^{(n)}$. Since f is holomorphic at z_0 we can write

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \psi(z)(z - z_0),$$

where $\psi(z) \rightarrow 0$ as $z \rightarrow z_0$. Since the constant $f(z_0)$ and the linear function $f'(z_0)(z - z_0)$ have primitives, we can integrate the above equality using Corollary 3.3 in the previous chapter, and obtain

$$(3) \quad \int_{T^{(n)}} f(z) dz = \int_{T^{(n)}} \psi(z)(z - z_0) dz.$$

Now z_0 belongs to the closure of the solid triangle $T^{(n)}$ and z to its boundary, so we must have $|z - z_0| \leq d^{(n)}$, and using (3) we get, by (iii) in Proposition 3.1 of the previous chapter, the estimate

$$\left| \int_{T^{(n)}} f(z) dz \right| \leq \epsilon_n d^{(n)} p^{(n)},$$

where $\epsilon_n = \sup_{z \in T^{(n)}} |\psi(z)| \rightarrow 0$ as $n \rightarrow \infty$. Therefore

$$\left| \int_{T^{(n)}} f(z) dz \right| \leq \epsilon_n 4^{-n} d^{(0)} p^{(0)},$$

which yields our final estimate

$$\left| \int_{T^{(0)}} f(z) dz \right| \leq 4^n \left| \int_{T^{(n)}} f(z) dz \right| \leq \epsilon_n d^{(0)} p^{(0)}.$$

Letting $n \rightarrow \infty$ concludes the proof since $\epsilon_n \rightarrow 0$.

Corollary 1.2 *If f is holomorphic in an open set Ω that contains a rectangle R and its interior, then*

$$\int_R f(z) dz = 0.$$

This is immediate since we first choose an orientation as in Figure 2 and note that

$$\int_R f(z) dz = \int_{T_1} f(z) dz + \int_{T_2} f(z) dz.$$

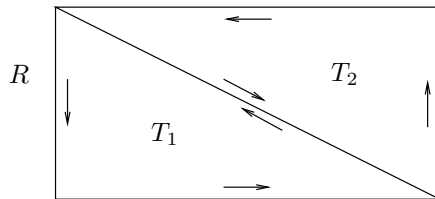


Figure 2. A rectangle as the union of two triangles

2 Local existence of primitives and Cauchy's theorem in a disc

We first prove the existence of primitives in a disc as a consequence of Goursat's theorem.

Theorem 2.1 *A holomorphic function in an open disc has a primitive in that disc.*

Proof. After a translation, we may assume without loss of generality that the disc, say D , is centered at the origin. Given a point $z \in D$, consider the piecewise-smooth curve that joins 0 to z first by moving in the horizontal direction from 0 to $\tilde{z}$ where $\tilde{z} = \operatorname{Re}(z)$, and then in the vertical direction from $\tilde{z}$ to z . We choose the orientation from 0 to z , and denote this polygonal line (which consists of at most two segments) by γ_z , as shown on Figure 3.

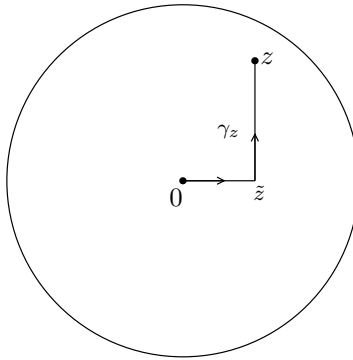


Figure 3. The polygonal line γ_z

Define

$$F(z) = \int_{\gamma_z} f(w) dw.$$

The choice of γ_z gives an unambiguous definition of the function $F(z)$. We contend that F is holomorphic in D and $F'(z) = f(z)$. To prove this, fix $z \in D$ and let $h \in \mathbb{C}$ be so small that $z + h$ also belongs to the disc. Now consider the difference

$$F(z + h) - F(z) = \int_{\gamma_{z+h}} f(w) dw - \int_{\gamma_z} f(w) dw.$$